

# ECE 332 — Homework 4: Generated Answers from HW4 Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

**Symbol reference** (used throughout):

- $l$  — antenna physical length (m)
- $\lambda = c/f$  — wavelength (m);  $c = 3 \times 10^8$  m/s
- $I_0$  — peak current at antenna feed (A)
- $R$  — distance from antenna to observation point (m)
- $\theta$  — angle from antenna axis (rad or deg)
- $\eta_0 = 120\pi \approx 377 \Omega$  — intrinsic impedance of free space
- $k = 2\pi/\lambda$  — wavenumber (rad/m)
- $S(R, \theta)$  — average power density radiated (W/m<sup>2</sup>)
- $F(\theta) = S(\theta)/S_{\max}$  — normalized radiation intensity (unitless)
- $\beta$  — half-power beamwidth (rad or deg)
- $\Omega_p$  — pattern solid angle (sr)
- $D = 4\pi/\Omega_p$  — directivity (unitless, often dB)
- $A_e = \lambda^2 D/(4\pi)$  — effective aperture area (m<sup>2</sup>)
- $A_p$  — physical cross-section area (m<sup>2</sup>)
- $G_t, G_r$  — antenna gains (linear)
- $P_t, P_r$  — transmitted / received power (W)

**Key shortcuts:** Hertzian dipole ( $l/\lambda < 1/50$ )  $\Rightarrow D = 1.5$ . Half-wave dipole ( $l = \lambda/2$ )  $\Rightarrow D = 1.64$ ,  $R_{\text{rad}} = 73 \Omega$ . 3 dB  $\Rightarrow$  linear gain of 2.

## Problem 1 — Hertzian Dipole, Power Density at 5 km

8 pts

**Setup:** A 1-m long dipole excited by a 1 MHz current with amplitude  $I_0 = 12$  A. Find the average power density radiated at  $R = 5$  km,  $\theta = 45^\circ$ .

### DIPOLE TYPE BY LENGTH

| Compute first $\lambda = c/f$ , then ratio $l/\lambda$ |                  |                                                                |
|--------------------------------------------------------|------------------|----------------------------------------------------------------|
| $l/\lambda$                                            | Type             | Use formula                                                    |
| $< 1/50$                                               | Hertzian (short) | HERTZIAN $S(R, \theta)$ , $D = 1.5$                            |
| $= 1/4$                                                | Quarter-wave     | ARB formula, $\pi l/\lambda = \pi/4$                           |
| $= 1/2$                                                | Half-wave        | half-wave short-cut, $D = 1.64$ , $R_{\text{rad}} = 73 \Omega$ |
| $= 1$                                                  | Full-wave        | ARB, $D = 2.41$ , $R_{\text{rad}} = 199 \Omega$                |
| other                                                  | Arbitrary        | ARB formula                                                    |

If  $l/\lambda < 1/50$  STOP and use Hertzian. Do NOT plug small  $l$  into the arbitrary formula — wrong shape.

### HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small  $l \ll \lambda$

$$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi} \frac{e^{-jkR}}{R} \sin \theta, \quad \vec{H}_\phi = \vec{E}_\theta / \eta_0$$

Power density  $l/\lambda < 1/50$ ; far field

$$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\max} \sin^2 \theta \text{ (W/m}^2\text{)}$$

$k = 2\pi/\lambda$  (rad/m);  $I_0$  peak current (A);  $l$  length (m);  $R$  obs. dist (m);  $\eta_0 = 120\pi \approx 377 \Omega$ .

Max direction broadside  $\theta_{\max} = \pi/2$

Total radiated power integrate over  $4\pi$

$$P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left( \frac{I_0 l}{\lambda} \right)^2 \text{ (W)}$$

Directivity exact  $D_{\text{Hertz}} = 1.5$  (1.76 dB)

## COMMON DIPOLE PARAMETERS

| Type          | $D$  | $D_{\text{dB}}$ | $R_{\text{rad}} (\Omega)$ |
|---------------|------|-----------------|---------------------------|
| Isotropic     | 1    | 0               | —                         |
| Hertzian      | 1.5  | 1.76            | $80\pi^2(l/\lambda)^2$    |
| Half-wave     | 1.64 | 2.15            | 73                        |
| $l = \lambda$ | 2.41 | 3.82            | 199                       |

*Half-wave is the default “dipole antenna” unless stated otherwise.*

**Step 1 — Classify the dipole length.** Compute the wavelength to compare with  $l$ :

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^6} = 300 \text{ m}$$

$$\frac{l}{\lambda} = \frac{1}{300} < \frac{1}{50} \Rightarrow \text{Hertzian dipole}$$

The DIPOLE TYPE BY LENGTH section says short ( $l/\lambda < 1/50$ ) means we can use the Hertzian  $S(R, \theta)$  formula directly.

**Step 2 — Apply the Hertzian power density formula.** From HERTZIAN DIPOLE —  $S(R, \theta)$ :

$$S(R, \theta) = \left( \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \right) \sin^2 \theta$$

**Step 3 — Plug in numbers.**  $\eta_0 = 120\pi$ ,  $k = 2\pi/\lambda = 2\pi/300 \text{ rad/m}$ ,  $I_0 = 12 \text{ A}$ ,  $l = 1 \text{ m}$ ,  $R = 5000 \text{ m}$ ,  $\theta = \pi/4$ :

$$S = \frac{(120\pi)(2\pi/300)^2(12)^2(1)^2}{32\pi^2(5 \times 10^3)^2} \sin^2(\pi/4)$$

$$\text{Numerator: } (120\pi)(4\pi^2/9 \times 10^4)(144) = (120\pi)(144)(4\pi^2)/(9 \times 10^4)$$

$$= 120 \cdot 144 \cdot 4 \cdot \pi^3 / (9 \times 10^4) = 69120\pi^3 / (9 \times 10^4)$$

$$\text{Denominator: } 32\pi^2 \cdot 2.5 \times 10^7. \text{ And } \sin^2(\pi/4) = 1/2.$$

Working it out cleanly:

$$S(5 \text{ km}, 45^\circ) = 1.51 \times 10^{-9} \text{ W/m}^2$$

*Sanity:* a 12 A current at 1 MHz across a 1-m antenna at 5 km radiates 1.5 nW/m<sup>2</sup>. The  $1/R^2$  falloff dominates: most of the radiated power scatters into space, only a tiny fraction reaches any one observation point.

## Problem 2 — Beam Parameters from Normalized Pattern

15 pts

**Setup:**  $F(\theta) = \exp(-20\theta^2)$  for  $0 \leq \theta \leq \pi$  ( $\theta$  in radians). Find: (a) half-power beamwidth  $\beta$ , (b) pattern solid angle  $\Omega_p$ , (c) directivity  $D$ .

BEAM PARAMETERS —  $\beta$ ,  $\Omega_p$ ,  $D$ 

**Step 1 — Normalize** *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

**Step 2 — HPBW**  $\beta$  *solve  $F(\theta_{1/2}) = 0.5$ , then  $\beta = 2\theta_{1/2}$  for symmetric beams*

*Gaussian shortcut:  $F = e^{-a\theta^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$ .*

**Step 3 — Pattern solid angle  $\Omega_p$**  *units: sr*

$$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$$

*No  $\phi$  dep.:  $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$ .*

*Most integrals have NO closed form. Use Wolfram / numerical.*

*Wolfram:  $2\pi \text{Integrate}[F[t]*\text{Sin}[t], \{t, 0, \pi\}]$*

**Step 4 — Directivity  $D$**

$$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \quad D_{\text{dB}} = 10 \log_{10} D$$

*Alt. with two orthogonal HPBWs:  $D \approx 4\pi/(\beta_x \beta_y)$ .*

**Step 5 — Gain (if asked)**  $\xi = \text{rad. eff.}, 0 < \xi \leq 1$

$$G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$$

*Lossless / ideal antenna  $\Rightarrow G = D$ .*

(a) **Half-power beamwidth  $\beta$  (5 pts).** From BEAM PARAMETERS Step 2, the original cheat sheet equations:

$$F(\theta_{1/2}) = 0.5 \Rightarrow \text{solve for } \theta_{1/2}$$

$$\beta = 2\theta_{1/2} \text{ (symmetric beams)}$$

Plug in  $F(\theta) = \exp(-20\theta^2)$ :

$$\exp(-20\theta_{1/2}^2) = 0.5 \Rightarrow -20\theta_{1/2}^2 = \ln(0.5)$$

$$\theta_{1/2} = \sqrt{-\ln(0.5)/20} = \sqrt{0.693/20} = 0.186 \text{ rad}$$

That's the half-angle from the peak to one  $-3$  dB point. Since the beam is symmetric about  $\theta = 0$ , the *full half-power beamwidth* is twice that:

$$\beta = 2\theta_{1/2} = 0.372 \text{ rad} = 21.31^\circ$$

**(b) Pattern solid angle  $\Omega_p$  (5 pts).** From BEAM PARAMETERS Step 3:

$$\Omega_p = \iint_{4\pi} F(\theta) \sin \theta \, d\theta \, d\phi = \int_0^{2\pi} \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta \, d\phi$$

$F$  has no  $\phi$  dependence, so  $\int_0^{2\pi} d\phi = 2\pi$ :

$$\Omega_p = 2\pi \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta$$

This integral has no closed form; evaluate numerically on the TI-36X Pro.

**TI-36X Pro keystrokes (definite integral):**

1. Press `[2nd] → [d/dx]` ( $\rightarrow$  definite-integral template appears, shown as  $\int_{\square}^{\square} \square \, d\square$ ).
2. **Lower limit:** type 0, then  $\rightarrow$  arrow to the upper limit.
3. **Upper limit:** press `[ $\pi$ ]`, then arrow into the integrand box.
4. **Integrand:** type `e^(-20×x^2)×sin(x)` (use `[e^x]` for the exp, `[x^2]` for the square, `[sin]` for the sine).
5. **Variable:** arrow to the  $d\square$  slot, type `x`.
6. Press `[enter]`  $\rightarrow$  calculator returns  $\approx 0.024792$ .
7. Multiply by  $2\pi$ : `×[ $\pi$ ]×2 [enter]`  $\rightarrow \approx 0.15578$ .

$$\Omega_p \approx 0.1558 \text{ sr}$$

Keep the unrounded calculator value for part (c). Textbook rounds  $\Omega_p$  to 0.156 first and gets  $D = 80.55$ ; using the full precision gives  $D \approx 80.67$ . Make sure the calculator is in RADIANS mode (`[mode] → RAD`) since  $\theta$  in the exponent is bare — degree mode will give nonsense.

**(c) Directivity  $D$  (5 pts).** From BEAM PARAMETERS Step 4:

$$D = \frac{4\pi}{\Omega_p} = \frac{4\pi}{0.15578}$$

$$D = 80.67$$

Using textbook's rounded  $\Omega_p = 0.156$ :  $D = 4\pi/0.156 = 80.55$ . Use the unrounded value to avoid this rounding error.

*Sanity:*  $D \approx 80$  is highly directive (much narrower than half-wave dipole's 1.64). The narrow Gaussian-like beam ( $\beta \approx 21^\circ$ ) concentrates radiation into a small solid angle, so directivity is large.

Problem 3 —  $\lambda/4$  Dipole Radiation Pattern

15 pts

**Setup:** Dipole of length  $l = \lambda/4$ . Find: (a) directions of maximum radiation, (b) expression for  $S_{\max}$ , (c) plot of normalized pattern  $F(\theta)$ .

ARBITRARY-LENGTH DIPOLE —  $S(\theta)$ Power density at  $R$  any  $l$ 

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[ \frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

Normalized pattern *dimensionless*  $F(\theta) = S(\theta)/S_{\max}$ Quarter-wave  $l = \lambda/4$   $\pi l/\lambda = \pi/4$ ,  $\cos(\pi/4) = \sqrt{2}/2$ 

$$\frac{S(\theta)}{S_0} = \left[ \frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$$

Half-wave  $l = \lambda/2$   $\pi l/\lambda = \pi/2$ , *shortcut form*

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta}$$

 $D = 1.64$ ,  $R_{\text{rad}} = 73 \Omega$ ,  $\theta_{\max} = \pi/2$ .

## COMMON DIPOLE PARAMETERS

| Type          | $D$  | $D_{\text{dB}}$ | $R_{\text{rad}} (\Omega)$ |
|---------------|------|-----------------|---------------------------|
| Isotropic     | 1    | 0               | —                         |
| Hertzian      | 1.5  | 1.76            | $80\pi^2 (l/\lambda)^2$   |
| Half-wave     | 1.64 | 2.15            | 73                        |
| $l = \lambda$ | 2.41 | 3.82            | 199                       |

Half-wave is the default “dipole antenna” unless stated otherwise.

**General arbitrary-length dipole pattern.** From ARBITRARY-LENGTH DIPOLE —  $S(\theta)$ :

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[ \frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

Plug in  $l = \lambda/4$ , so  $\pi l/\lambda = \pi/4$ :

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[ \frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \cos(\pi/4)}{\sin \theta} \right]^2 = \frac{15I_0^2}{\pi R^2} \left[ \frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$$

Let  $S_0 = 15I_0^2/(\pi R^2)$  be the constant prefactor.

**(a) Direction of maximum radiation (5 pts).** Three-step argument: (i) endpoints via L'Hôpital, (ii) symmetry about  $\pi/2$ , (iii) sampling to confirm single peak.

*Step (i) — endpoints  $\theta = 0, \pi$  via L'Hôpital.* Inside the squared bracket:  $f(\theta) = \cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}$  over  $g(\theta) = \sin \theta$ .

At  $\theta = 0$ :  $f(0) = \cos(\pi/4) - \sqrt{2}/2 = 0$  and  $g(0) = 0 \Rightarrow 0/0$  form. Apply L'Hôpital:

$$f'(\theta) = \frac{\pi}{4} \sin\left(\frac{\pi}{4} \cos \theta\right) \sin \theta, \quad g'(\theta) = \cos \theta$$

$$\lim_{\theta \rightarrow 0} \frac{f'(\theta)}{g'(\theta)} = \frac{(\pi/4) \sin(\pi/4) \cdot 0}{1} = 0$$

So  $S(0)/S_0 = 0^2 = 0$ . Same algebra at  $\theta = \pi$  (with  $g'(\pi) = -1$ ) gives  $S(\pi) = 0$ .

*Step (ii) — symmetry about  $\theta = \pi/2$ .* Substitute  $\theta \rightarrow \pi - \theta$ :  $\cos(\pi - \theta) = -\cos \theta$ , but  $\cos$  is even so  $\cos((\pi/4) \cos(\pi - \theta)) = \cos((\pi/4) \cos \theta)$ . And  $\sin(\pi - \theta) = \sin \theta$ . So  $S(\theta) = S(\pi - \theta)$ , pattern mirrors about  $\pi/2$ .

*Step (iii) — sample values to confirm single peak (not M-shape).* Compute  $S(\theta)/S_0$  for a few angles:

| $\theta$ | 30°   | 60°   | 90°          |
|----------|-------|-------|--------------|
| $S/S_0$  | 0.020 | 0.063 | <b>0.086</b> |

Monotone rise to  $90^\circ$ , then symmetric fall. Single peak at the center.

$$\theta_{\max} = \pi/2 = 90^\circ$$

(broadside to the dipole axis, same as half-wave dipole.)

**(b)  $S_{\max}$  (5 pts).** Evaluate  $S(\pi/2)$ :

$$S_{\max} = S(\pi/2) = \frac{15I_0^2}{\pi R^2} \left[ \frac{\cos(\frac{\pi}{4} \cdot 0) - \frac{\sqrt{2}}{2}}{\sin(\pi/2)} \right]^2 = \frac{15I_0^2}{\pi R^2} \left( 1 - \frac{\sqrt{2}}{2} \right)^2$$

Numerically  $(1 - \sqrt{2}/2)^2 = 0.0858$ :

$$S_{\max} = 0.0858 \frac{15I_0^2}{\pi R^2}$$

**(c) Normalized pattern  $F(\theta) = S(\theta)/S_{\max}$  (5 pts):**

$$F(\theta) = \frac{1}{0.0858} \left[ \frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$$

Bell-shaped curve peaking at  $\theta = \pi/2$  with  $F = 1$ , dropping to 0 at  $\theta = 0$  and  $\pi$ . Symmetric about  $\theta = \pi/2$ . Similar shape to half-wave dipole pattern but slightly narrower since the dipole is shorter.

*Sanity:* A  $\lambda/4$  dipole has a broadside pattern very close to a Hertzian dipole's  $\sin^2 \theta$  shape. As  $l \rightarrow 0$  the bracket approaches  $\sin \theta$  and  $S \propto \sin^2 \theta$  (Hertzian). Half-wave  $l = \lambda/2$  has slightly narrower beam than the  $\lambda/4$  case.

## Problem 4 — Effective Area vs. Physical Cross-section

5 pts

**Setup:** Half-wave dipole at  $f = 100$  MHz, wire diameter  $d = 2$  cm. Find effective area  $A_e$  and compare to physical cross-section  $A_p$ .

EFFECTIVE AREA  $A_e$

Definition
*antenna's RX cross section ( $m^2$ )*

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$$

Physical cross section
*wire dipole, diameter  $d$*

$$A_p = l d = \frac{\lambda}{2} d \text{ (half-wave)}$$

Inverse:  $D$  from  $A_e$

$$D = \frac{4\pi A_e}{\lambda^2} \text{ (unitless)}$$

*Thin wire:  $A_e \gg A_p$ . Antenna captures field over  $\sim \lambda$ .*

COMMON DIPOLE PARAMETERS

| Type          | $D$  | $D_{dB}$ | $R_{rad} \text{ (}\Omega\text{)}$ |
|---------------|------|----------|-----------------------------------|
| Isotropic     | 1    | 0        | —                                 |
| Hertzian      | 1.5  | 1.76     | $80\pi^2 (l/\lambda)^2$           |
| Half-wave     | 1.64 | 2.15     | 73                                |
| $l = \lambda$ | 2.41 | 3.82     | 199                               |

*Half-wave is the default “dipole antenna” unless stated otherwise.*

**Step 1 — Wavelength.**

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3 \text{ m}$$

**Step 2 — Directivity for half-wave dipole.** From COMMON DIPOLE PARAMETERS:  $D = 1.64$ .

**Step 3 — Effective area.** From EFFECTIVE AREA  $A_e$ :

$$A_e = \frac{\lambda^2 D}{4\pi} = \frac{(3)^2 (1.64)}{4\pi} = \frac{14.76}{4\pi}$$

$$A_e = 1.17 \text{ m}^2$$

**Step 4 — Physical cross-section.** A half-wave dipole has length  $l = \lambda/2$ . Treat as a rectangle of length  $l$  and width  $d$ :

$$A_p = l \cdot d = \frac{\lambda}{2} \cdot d = \frac{3}{2}(0.02) = 0.03 \text{ m}^2$$

**Step 5 — Ratio.**

$$\frac{A_e}{A_p} = \frac{1.17}{0.03} = 39.2$$

*Sanity:* The effective area is  $39\times$  larger than the physical cross-section. This is the key insight of antenna theory: a thin wire antenna captures energy from a much larger "effective" patch of the incoming wave than its physical wire would suggest, because the antenna interacts with the field over a region  $\sim \lambda$  around itself.

## Problem 5 — Friis Transmission, TV Broadcast

5 pts

**Setup:** Half-wave dipole TX, 1 kW at 50 MHz. RX home antenna with 3 dB gain, distance 30 km. Find received power.

## FRIIS TRANSMISSION FORMULA

Far-field link budget  $P_t$  tx,  $P_r$  rx, range  $R$ 

$$P_r = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 P_t \quad (\text{W})$$

 $G_t, G_r$  are linear gains;  $\lambda = c/f$ ;  $R$  in m;  $P_t, P_r$  in W.Solve for  $R$  max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \quad (\text{m})$$

Equivalent form using  $A_e$  recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss:  $L_{fs} = (4\pi R/\lambda)^2$ , so  $P_r = G_t G_r P_t / L_{fs}$ .Recipe 1.  $\lambda = c/f$ . 2. Convert dB gains to linear:  $G = 10^{G_{dB}/10}$ .3. If TX is “half-wave dipole” use  $G_t = 1.64$ . 4. Plug into Friis.Use linear  $G$ , not dB. Convert FIRST.

## dB ↔ LINEAR

$$G_{dB} = 10 \log_{10} G_{lin} \quad G_{lin} = 10^{G_{dB}/10}$$

| dB | linear | dB  | linear |
|----|--------|-----|--------|
| 0  | 1      | 10  | 10     |
| 3  | 2      | 13  | 20     |
| 6  | 4      | 20  | 100    |
| 7  | 5      | 30  | 1000   |
| -3 | 0.5    | -10 | 0.1    |

Memorize: 3 dB  $\Leftrightarrow \times 2$ , 10 dB  $\Leftrightarrow \times 10$ . Add dB = multiply linear.For E-field / amplitude use 20 log not 10 log (power  $\propto E^2$ ).

## Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{5 \times 10^7} = 6 \text{ m}$$

**Step 2 — Antenna gains.** TX is half-wave dipole, so  $G_t = 1.64$  (from COMMON DIPOLE PARAMETERS).

RX gain: 3 dB  $\rightarrow$  linear via  $G_{lin} = 10^{G_{dB}/10}$ :

$$G_r = 10^{3/10} = 2$$

(3 dB always =  $\times 2$ , this is the cleanest dB conversion to remember.)**Step 3 — Friis formula.** From FRIIS TRANSMISSION FORMULA:

$$P_r = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 P_t$$

**Step 4 — Plug in.**  $R = 30 \text{ km} = 3 \times 10^4 \text{ m}$ ,  $P_t = 10^3 \text{ W}$ :

$$P_r = (1.64)(2) \left( \frac{6}{4\pi(3 \times 10^4)} \right)^2 (10^3)$$

The path-loss factor:

$$\frac{\lambda}{4\pi R} = \frac{6}{4\pi \cdot 3 \times 10^4} = \frac{6}{3.77 \times 10^5} = 1.59 \times 10^{-5}$$

$$\left( \frac{\lambda}{4\pi R} \right)^2 = 2.53 \times 10^{-10}$$

$$P_r = (1.64)(2)(2.53 \times 10^{-10})(10^3) = (3.28)(2.53 \times 10^{-7})$$

$$P_r = 8.3 \times 10^{-7} \text{ W} = 0.83 \mu\text{W}$$

*Sanity:* 1 kW transmitted, 0.83  $\mu\text{W}$  received at 30 km. That’s a 90 dB drop, dominated by the  $1/R^2$  free-space path loss. Plenty of signal for a TV receiver (typical sensitivity is  $-100 \text{ dBm} = 10^{-13} \text{ W}$ ), so this link closes comfortably.

*Note on the solution PDF:* the official key wrote “(1.64)(3)” as a typo, but used  $G_r = 2$  in the actual arithmetic (since 3 dB  $\rightarrow$  linear = 2). The final answer  $8.3 \times 10^{-7} \text{ W}$  matches either way.